

Qatar-2b

R-1973

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_130213 · created 2026-08-07T00:06:12+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456336.95084 +/- 0.0003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.014
Transit depth (Rp/Rs)^2	2.29 +/- 0.43 [%]
Orbital Inclination (inc)	88.85 +/- 1.16
Airmass coefficient 1 (a1)	1.027 +/- 0.014
Airmass coefficient 2 (a2)	-0.014 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#1 - [57, 134]
Optimal Method	PSF photometry
Transit Duration (day)	0.0754 +/- 0.0015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.151 ± 0.014 vs 0.16327 (deviation 0.88σ)
Duration fit vs published	0.0754 d vs 0.0767 d (deviation 0.84σ)
Mid-time O–C	0.8 min
Depth SNR	10.8
Residual scatter	1.35 %

Light curve

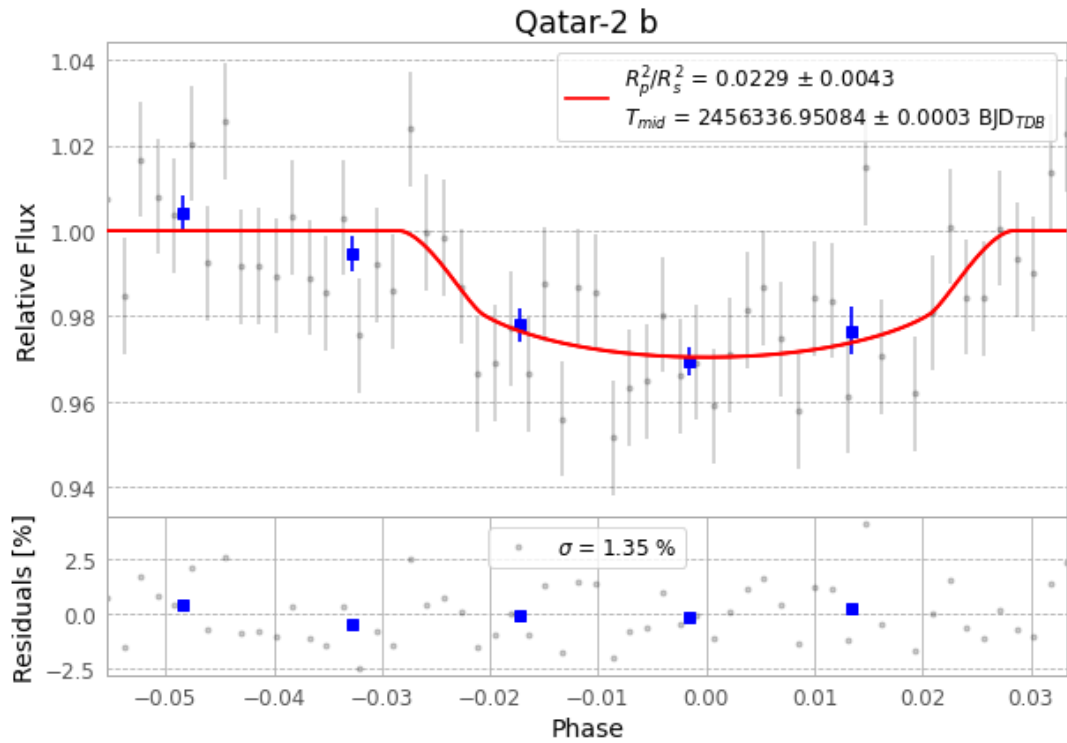


Figure 1 — FinalLightCurve_Qatar-2 b_2013-02-13.png (SHA-256 42b577be4954e697...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 277], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a8ef06de191aad1d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

58 FITS frames (38 MB), Qatar - 2130213085712.FITS → Qatar - 2130213114813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-130213.log (d7f2a6c2d2a5...), FinalLightCurve_Qatar-2 b_2013-02-13.png (42b577be4954...), FinalLightCurve_Qatar-2 b_2013-02-13.csv (8c26c9c06556...)

Compute resources

Wall time 107.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:06Z.